



AMITY UNIVERSITY
— **UTTAR PRADESH** —

DOMAIN: ENGINEERING AND TECHNOLOGY

INTERNSHIP-I

i. Student Details

Name	Samay Kumar
Enrollment Number	A2305325001
Programme	B.Tech – Information Technology
Year / Semester	2nd Year / 3rd Semester
Academic Session	2026–2027
Section	B.Tech IT
Institution	Amity School of Engineering & Technology, AUUP, Noida

ii. Faculty Guide

Name	Ms. Veenu Singh
Designation	Faculty Guide, ASET – Amity University, Noida

iii. Topic

Centralized vs Peer-to-Peer Open-Source VPN Architectures in Linux Environments

iv. Objective(s)

- To analyze and compare centralized VPN architectures (WireGuard) and peer-to-peer mesh VPN architectures (Headscale) deployed on Linux systems.
- To evaluate performance across key metrics: latency, throughput, CPU utilization, and packet loss using standard Linux benchmarking tools.
- To assess fault tolerance and resilience of each VPN architecture under simulated node and server failure conditions.
- To examine the security models, encryption mechanisms, trust assumptions, and infrastructure dependencies of each VPN solution.
- To design and implement a real-time performance visualization dashboard displaying live benchmark data, topology diagrams, and node health status.
- To produce a structured comparative analysis to inform practical VPN deployment decisions in Linux-based infrastructure.

v. Description

Virtual Private Networks (VPNs) are widely used for establishing secure communication across distributed systems and cloud-based environments. Different VPN architectures offer varying trade-offs in performance, scalability, and infrastructure dependency.

This study investigates two dominant VPN paradigms: the centralized (hub-and-spoke) model, where all encrypted traffic routes through a designated central server, and the peer-to-peer (mesh) model, where nodes establish direct encrypted tunnels with one another. The centralized side is represented by WireGuard in traditional server deployment mode. The peer-to-peer side is represented by Headscale. Two modern mesh VPN solutions widely adopted in open-source and enterprise Linux environments.

Testing is conducted on Ubuntu 22.04 LTS cloud virtual machines using standard Linux networking tools including TUN/TAP interfaces, WireGuard kernel modules, iproute2, and traffic control utilities. Benchmarking covers latency, throughput, CPU usage, and packet loss under normal and fault conditions. The project also includes development of a live monitoring dashboard built with Node.js, React, and Socket.IO, displaying real-time benchmark data, node connectivity status, and comparative metrics during test runs.

vi. Proposed Methodology

The study follows a five-module approach:

1. **Architectural Study** - Review of technical documentation and deployment models for WireGuard and Headscale-based networking. Study of centralized vs peer-to-peer coordination models, key exchange mechanisms, and routing behavior.
2. **Environment Setup and Deployment** - Provisioning of Ubuntu 22.04 LTS cloud VMs. Configuration of WireGuard in centralized hub-and-spoke mode and Headscale in peer-to-peer mesh mode. Verification of tunnel establishment, routing, and DNS resolution across nodes.
3. **Performance Benchmarking** - Execution of standardized benchmarks using ping, iperf3, and Linux monitoring tools. Tests cover latency under load, maximum throughput, CPU utilization per VPN type, and packet loss under congestion. Fault tolerance tested by simulating node shutdowns and observing recovery behavior.
4. **Dashboard Development** - Development of a lightweight real-time dashboard using Node.js (Express, Socket.IO) and React with Chart.js or Recharts. The dashboard streams live benchmark metrics, displays VPN topology, and shows node health status during active test runs.
5. **Comparative Analysis and Documentation** - Compilation of results into structured comparison tables across five dimensions: performance, scalability, fault tolerance, security model, and deployment complexity. Final observations drawn from empirical data rather than theoretical assumptions.

vii. Resources Required

Category	Details
Hardware	Personal computer / laptop – minimum 16 GB RAM, stable internet connection, multi-core CPU
Cloud Infrastructure	Ubuntu 22.04 LTS cloud virtual machines (Ubuntu-based VPS instances)
VPN Software	WireGuard (centralized), Headscale + Tailscale client (peer-to-peer mesh)
Benchmarking Tools	ping, iperf3, htop, top, wg, ip route, tc (traffic control)
Backend / Frontend	Node.js, Express, React, Tailwind CSS, Chart.js
Documentation	Git + GitHub

viii. Work Schedule

Duration: 1st June 2026 – 3rd July 2026 (5 Weeks)

Activity / Task	Wk 1 (1–7 Jun)	Wk 2 (8–14 Jun)	Wk 3 (15–21 Jun)	Wk 4 (22–28 Jun)	Wk 5 (29 Jun–3 Jul)
Topic finalization & synopsis submission	✓				
Literature review & theoretical study	✓	✓			
VM setup & VPN software deployment		✓	✓		
WireGuard & Headscale configuration and testing			✓	✓	
Benchmarking & fault tolerance tests				✓	
Dashboard development & live visualization				✓	✓
Report writing & plagiarism check					✓
Final submission & GitHub documentation					✓

References

- [1] J. A. Donenfeld, "WireGuard: Next Generation Kernel Network Tunnel," NDSS Symposium, 2017. Available: <https://www.wireguard.com/papers/wireguard.pdf>
- [2] Tailscale Inc., "How Tailscale Works," Tailscale Engineering Blog, 2020. Available: <https://tailscale.com/blog/how-tailscale-works>
- [3] The Linux Kernel Documentation, "WireGuard – Fast, Modern, Secure VPN Tunnel." Available: <https://www.kernel.org/doc/html/latest/networking/wireguard.html>
- [4] juanfont, "Headscale: An open source Tailscale control server," GitHub, 2021. Available: <https://github.com/juanfont/headscale>
- [5] J. Rosen, "A Comparison of WireGuard, IKEv2/IPsec, and OpenVPN," Mullvad VPN Blog, 2020. Available: <https://mullvad.net/en/blog/>
- [6] iproute2 Maintainers, "ip-rule – routing policy database management," Linux man-pages. Available: <https://man7.org/linux/man-pages/man8/ip-rule.8.html>

Student Signature:

Samay Kumar | A2305325001

Date: _____

**Faculty Guide Signature:**

Ms. Veenu Singh

Date: _____

